258- Add Digits

~~Algo~~ Input : 38 Output : 2

Process to solve

Sum of digits = 3+8=11

Sum of digits = 1+1=2

Output = 2

digit processing while number >=10

Code

```
Class Solution {
    public int addDigits(int num) {
        while (num >= 10) {          → if false, it is already a single digit.
            int sum = 0;
            while (num > 0) {
                int digit = num % 10;     // extract the last digit
                sum += digit;             // add that digit to sum
                num = num / 10;           // remove the last digit
            }
            num = sum;
        }
        return num;
    }
}
```

digit processing and calculating Sum

Time Complexity - O(log n)

Space Complexity - O(1)

Most efficient way

```
class Solution {
    public int addDigits(int num) {
        if (num == 0)
            return 0;
        return (num % 9 == 0) ? 9 : (num % 9);
    }
}
```

Description | Accepted | Editorial | Solutions | Submissions

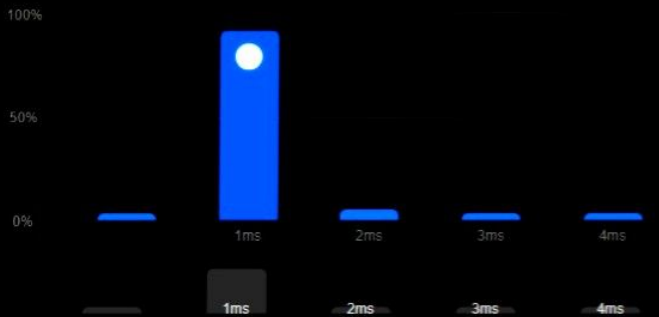
All Submissions

Runtime

1 ms | Beats **98.46%**

Memory

42.44 MB | Beats 73.41%



Code

Java | Auto

```java
class Solution {
    public int addDigits(int num) {

        if(num == 0) return 0;
        return (num % 9 == 0) ? 9 : (num % 9);
    }
}
```

Saved

Ln 1, Col 17

Testcase | Test Result

Accepted Runtime: 0 ms

Case 1 | Case 2

Input

num =

38

Output